School of Science, Engineering and Technology

Mechanical Design

MIET2510

Submitted By:

Vu Tran Khoi Nguyen – s3978262

Date Submitted: 27/09/2024

Contents

Executive Summary 3

1. Introduction 3

1.1 Background 3

1.2 Objectives 3

1.3 Scope 3

1.4 Report Structure 4

2 Design Methodology 4

2.1 Shaft design 4

2.1.1 Assumption 4

2.1.2 Calculation 5

2.2 Bearings design 7

2.2.1 Assumption 7

2.2.2 Calculation 9

2.3 Key – keyway design 11

2.3.1 Assumption 11

2.3.2 Calculation 11

2.4 Bolt design 11

2.4.1 Assumption 11

2.4.2 Calculation 11

3 Results 12

3.1 Shaft arrangement 12

3.2 Component design result 12

4. Discussion 13

4.1 Design implications: 13

4.2 Comparisons with standard: 13

4.3 Limitations: 14

4.4 Recommendation: 14

5. Conclusions 14

6. Appendix 14

6.1 Citation 14

6.2 2D Drawing: 15

Executive Summary

This project focuses on designing the main drive shaft, bearings, keyed joint, and shaft-to-turbine connections for a vertical axis hydrogenerator. The design process involves selecting appropriate mechanical components—such as the shaft, bearings, keyed joint, and fasteners—based on criteria like weight, durability, and dimensional compatibility, all while adhering to specific constraints through detailed stress and torque calculations.

The goal is to create a conceptual design for a vertical axis hydrogenerator that operates at a nominal speed of 115 RPM, generates up to 200 kW of power, and weighs between 600 and 1050 kg. The design aims to achieve a reliability of 98% with a safety factor of 2. [1]

1. Introduction

1.1 Background

Hydrogenerator are devices used to generate electrical power from the gravitational potential energy gained from water flow. Therefore, reduce reliance on fossil fuel and natural resources, while this device generates renewable energy (hydropower).

The design of this hydrogenerator need to consider design of bearings, shaft, bolt, key-keyways that can withstand stresses while keeping high reliability and a good durability. The optimization of weight, size and cost effectiveness are also critical factors of design process.

This project is relevant in the context of increasing demand of renewable energy. Efficient and effective hydrogenator will greatly enhance power generating capability.

1.2 Objectives

The main goal is to create an effective hydrogenator. The design needs to address the calculation / selection of these components with a safety factor of 2:

- Shaft
- Key – keyways to transfers torque from shaft to rotor
- Bearings which are thrust bearing (support the turbine vertically), guide bearing (support the turbine horizontally)
- Bolt is used for shaft to turbine connection.

1.3 Scope

Design specifications:

- Weight and size:
    - Rotor assembly estimated 500 - 800kg
    - Turbine estimated 500 - 800kg
    - Total length not exceeding 5 metres.
- Material selection: Based on design.
- Shaft arrangement: from upper guide bearing at the top to rotor to thrust bearing to lower guide bearing second bearing with key connections from rotor to shaft and bolt connection from shaft to turbine.
- Bearing: selection of upper guide bearing, lower guide bearing and thrust bearing.
- Component height: not exceed 5 meters.

Exclusion:

- Design of rotor, stator.
- Design of turbine, wicket gates.
- Detail stress analysis and system motion studies.

1.4 Report Structure

First, calculate the dimensions of the shaft. Once the shaft dimensions are determined, select the appropriate bearings based on the shaft's diameter. Finally, design the connections, including keyways and bolts, which are crucial for the assembly.

2 Design Methodology

2.1 Shaft design

First, the dimensions of the shaft need to be calculated before other components because it is the backbone of the design. The proper bearing diameter cannot be chosen without knowing the shaft diameter, and the same applies to the connections (key – keyways & bolt), which are purely based on the shaft design. Shaft arrangement also considered in this part.

2.1.1 Assumption

Safety factor $n_s = 2$ as stated in the project requirement.

For the shaft, the material is **AISI 316 Stainless Steel** so Yield strength $S_y = 205\ MPa$ and density $D = 8000 kg/m^3$[2]. This material is chosen for the shaft because 316 Stainless Steel offers excellent resistance to corrosion, especially in environments exposed to water,

including saltwater. This ensures the shaft remains durable and maintains its integrity over time.

The water is assumed to have density of $1000\frac{kg}{m^3}$ while this is the standard for water density. In perfect condition, water flow vertically to the turbine with angle of contact is 90^0, but due to dynamic environment of water flow itself and wicket gates design might have some tolerances. Angle of water flow into turbine is assumed to be 85^0.

Assuming that radius of the turbine is 0.8m because stated height for the turbine in the report ranging from 0.7 - 0.9m so median height 0.8m is used. Turbine height is assumed to be 0.8m height, with aspect ratio of 1:1 good balance between turbine height and span of the blades can be maintained.

Water column will have the height $h = 1 + 0.8 = 1.8m$ which is just right below lower guide bearing.

For weight optimization, this project considers the scenario where the weight of the rotating components is at its minimum. The typical weight range for the rotor is 200kg to 300kg, and for the turbine, it is 300kg to 400kg. Therefore, the median values of 250kg for the rotor and 350kg for the turbine have been selected.

Given that the height of the turbine is already 0.8 meters, and the total system height must not exceed 5 meters, the shaft height has been optimized to 4.2 meters. This maximizes the available space, facilitating easier placement of components so length of shaft is $L = 5 - 0.8 = 4.2m$.

The rotor assembly has a height range of 0.6 to 0.85 meters. Therefore, a median height of 0.725 meters has been selected

2.1.2 Calculation

Torque can be found as:

$$T = \frac{P}{\omega} = \frac{200 * 10^3}{115 * \frac{2\pi}{60}} = 16607.472Nm$$

Water force can be found using:

$w_{water} = \mathrm{h} * \rho * \pi * {r_{turbine}}^2 * \mathrm{g} = 1.8 * 1000 * \pi * 1^2 * 9.81 = 55474.243N$ [3]

Weight of rotor:

$w_{rotor} = 250 * 9.81 = 2452.5N$

Weight of turbine:

$w_{turbine} = 350 * 9.81 = 3433.5N$

Weight of the shaft can be express as:

$$w_{shaft} = \pi * \frac{{d_{shaft}}^2}{4} * L * density * g = \pi * \frac{{d_{shaft}}^2}{4} * 4.2 * 8000 * 9.81$$
$$= 258879.801{d_{shaft}}^2 N$$

Axial loading is sum of weight of rotor, turbine, water and shaft itself

$$P_{axial} = w_{water} * \sin(\alpha) + w_{turbine} + w_{rotor} + w_{shaft}$$
$$= 2452.5 + 3433.5 + 258879.801{d_{shaft}}^2 + 55474.243 * \sin(85)$$
$$= 61149.147 + 258879.801{d_{shaft}}^2$$

Radial load acts on the system:

$P_{radial} = w_{water} * \cos(\alpha) = 55474.243 * \cos(85) = 4834.899N$

Since guide bearings are supposed to mainly take radial load:

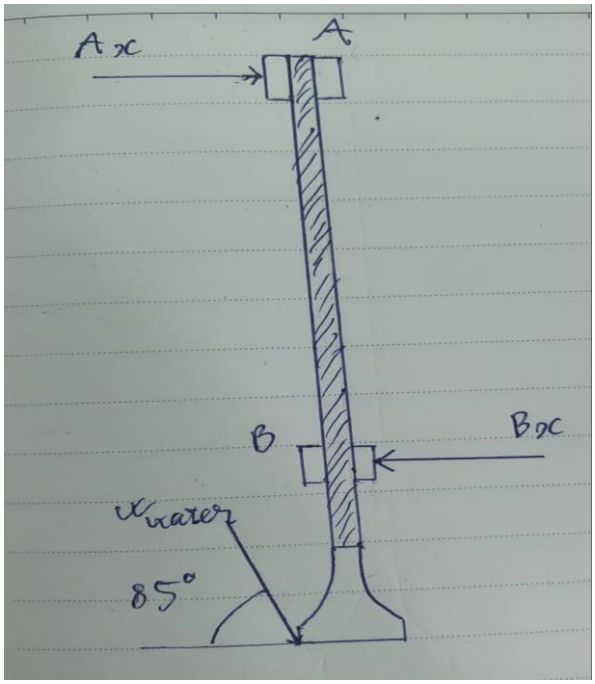


Figure 1: Free body diagram of the system

Using the equation of equilibrium:

- Therefore, sum of all force horizontally:

$$\sum F_x = 0 \rightarrow A_x + B_x - P_{radial} = 0 \rightarrow -A_x + B_x - 4834.899 = 0$$

- Sum of moment at point A (upper guide bearing):

$$\sum M_A = 0 \rightarrow B_x * 3.2 - P_{radial} * 5 = 0 \rightarrow B_x * 3.2 - 4834.899 * 5 = 0$$

Hence $\begin{cases} A_x = 2719.631N \\ B_x = 7554.529N \end{cases}$

Maximum bending moment appears at lower guide bearing: $M_{max} = A_x * 3.2 = 8702.819N$

Diameter of the shaft is calculated by substituting bending moment M, total axial force P, torque T, yield strength and safety factor to the equation below:

$$\frac{4}{\pi d^3}\sqrt{(8\mathrm{M} + \mathrm{P}d)^2 + 48\mathrm{T}^2} = \frac{\mathrm{S_y}}{\mathrm{n_s}}$$

Therefore:

$$\frac{4}{\pi {d_{shaft}}^3}\sqrt{\left(8 * 8702.819 + \left(61149.147 + 258879.801{d_{shaft}}^2\right)d_{shaft}\right)^2 + 48 * 16607.472^2} - \frac{205 * 10^6}{2} = 0$$

Using DESMOS to graph the equation above, the minimum allowable diameter of the shaft is $d_{shaft} = 0.1198m$ [4]

2.2 Bearings design

Based on the shaft diameter and the load acting on the bearing, the appropriate bearings can then be selected.

2.2.1 Assumption

Bearings type is selected from the following criteria.

Table 1: Guide bearing selection

Criteria	Ball Bearings	Roller Bearings	Selection
Cost-effectiveness	Generally cheaper and easier to maintain	More expensive due to complex manufacturing	**Ball Bearings**

Support Axial Load	Can handle moderate axial and radial loads[1]	Can handle high axial and radial loads[3]	**Ball Bearings** (sufficient for moderate loads, cost-effective)
Speed	Suitable for high-speed applications	Better for lower-speed applications	**Ball Bearings** (ideal for 115 RPM, which is moderate speed)
Weight	Lighter, reducing overall system weight[1]	Heavier, increasing system weight	**Ball Bearings**
Radial Load	Sufficient for moderate radial loads	Superior for high radial loads but may be over-engineer for this application	**Ball Bearings** (more cost-effective)

Ball bearings are the clear choice for the two guide bearings therefore exponent of life equation $a = 3$ is used.

For thrust bearing, roller bearing is used while:

Table 2: Thrust bearing selection

Criteria	Ball bearings	Roller bearings	Selection
Cost-effectiveness	Cheap	Expensive	**Ball bearings**
Support axial load	Cannot handle great axial load for this case	Can withstand great axial load	**Roller bearings (**great choice for thrust bearing)
Speed	High speed	Moderate speed	**Ball bearings**
Weight	light	heavy	**Ball bearings**
Support radial load	Can handle moderate radial load	Can handle great radial load	**Roller bearings**

Although ball bearings meet more criteria, they cannot handle the axial load, which is crucial for a thrust bearing. Therefore, roller bearings are necessary for this application, therefore exponent of life equation $a = \frac{10}{3}$.

Assume the hydrogenator works under moderate shock because:

- Water flow can vary due to change in water stream conditions → experiences fluctuations in load. This condition is more likely greater than light shock but not as heavy as heavy shock.
- Selecting bearing which handles moderate shock, in this case will make sure the design work successfully without being over-engineer.

For these reasons, moderate service shock $K_s = 1.3$ is selected for thrust bearing, $K_s = 2.0$ is selected for ball bearing. [5]

While guide bearings can handle some axial forces for added safety and reliability, their primary function is to manage radial loads. Consequently, the radial factor for these bearings is 1, and the thrust factor is 0.

On the other hand, thrust bearings, although capable of withstanding significant radial loads, are primarily designed to handle axial loading in this project. Therefore, the radial factor for these bearings is 0, and the thrust factor is 1.

All bearings will be mounted on the shaft, causing the inner rings to rotate. Therefore, a rotational factor of V= 1.0 for inner ring will be applied.

2.2.2 Calculation

Axial load for thrust bearing

$$F_{axial\ thrust} = P_{axial} = 61149.147 + 258879.801{d_{shaft}}^2$$
$$= 61149.147 + 258879.801 * 0.1198^2 = 64864.600N$$

Radial load for lower guide bearing is: $B_x = 7554.529N$

Radial load for upper guide bearing is: $A_x = 2719.631N$

Load acts on bearing can be calculated as: $P = K_s(XVF_r + YF_a)$

Load acts on the lower guide bearing can be calculate as

$$P = K_s(XVF_r + YF_a) = 2.0 * (1 * 1 * 7554.529 + 0 * 0) = 2 * 1 * 1 * 7554.529$$
$$= 15109.058N$$

Load acts on the upper guide bearing can be calculated as:

$$P = K_s(XVF_r + YF_a) = 2.0 * (1 * 1 * 2719.631 + 0 * 0) = 1.7 * 1 * 1 * 2719.631$$
$$= 5439.262N$$

Load acts on the thrust bearing can be calculated as:

$$P = K_s(XVF_r + YF_a) = 1.3 * (1 * 0 * 0 + 1 * 64864.600) = 84323.98N$$

Durability of the bearing $L_{10} = 8 * 365 * 24 = 70080$ hours

Life adjustment factor $K_r = 0.32$ for reliability of 98%

Dynamic load of bearing can be obtained from substituting load acts on bearing P, exponent of life, durability L_{10} and nominal speed n into the equation: $L_{10} = K_r * \frac{10^6}{60n}\left(\frac{C}{P}\right)^a$

Dynamic load of upper guide bearing:

$$70080 = 0.32 * \frac{10^6}{60 * 115}\left(\frac{C}{5439.262}\right)^3 \rightarrow C = 62417.214N$$

For the upper guide bearing, deep groove ball bearing, model 16026 is chosen, with bore diameter 130mm and dynamic load rating $C = 83.2$ kN. [6]

Dynamic load of lower guide bearing:

$$70080 = 0.32 * \frac{10^6}{60 * 115}\left(\frac{C}{15109.058}\right)^3 \rightarrow C = 173381.115N$$

For lower guide bearing, deep-groove ball bearing, model 6326 is chosen. Dynamic load and bore diameter of this model is 229kN and 130mm respectively. [6].

Dynamic load of thrust bearing:

$$70080 = 0.32 * \frac{10^6}{60 * 115}\left(\frac{C}{84323.98}\right)^{\frac{10}{3}} \rightarrow C = 758121.928N$$

Spherical roller thrust bearing with stamped steel cage model 29326 E is the most suitable option for thrust bearing with dynamic load rating 865 kN and bore diameter 130mm.[6]

Since the shaft diameter is already 119.8m therefore, bore diameter of bearings must have clearances for easy assembly which is $130 - d_{shaft} = 130 - 119.8 = 10.2mm$

2.3 Key – keyway design

2.3.1 Assumption

Material used for key design is **AISI 316 stainless steel** same as shaft's material, while using the same material ensures that the key and the shaft have similar strength and hardness. This compatibility helps prevent key failures due to material mismatch so yield strength $S_y = 205\ MPa$.

Safety factor $n = 2$ as stated in the project requirement.

The shaft $d_{shaft} = 119.8mm$ (in the range 111mm – 130mm):

➔ Width $w = 32mm$ is selected for key – keyway. [7]

➔ For key and keyway depth are 18mm and 7.4mm respectively. [7]

2.3.2 Calculation

Force at shaft surface can be found $F = \frac{T}{r} = \frac{16607.472}{0.1198/2} = 277253.288N$

By substituting force at shaft surface F, safety factor n, key's yield strength and width of key. Key and keyway length can be obtained via: $L = \frac{2Fn}{S_y w} = 0.169m$

2.4 Bolt design

2.4.1 Assumption

This design employs 10 evenly spaced bolts, each positioned 0.4 meters from the center of the shaft. This configuration ensures balanced load distribution, minimizing the risk of uneven stress. By spacing the bolts at a radius of 0.4 meters from the shaft center, the design maximizes torque transmission efficiency.

2.4.2 Calculation

Normal force load that each bolt needs to withstand to carry the turbine:

$$\sigma = \frac{w_{turbine}}{number\ of\ bolt * A} = \frac{350 * 9.81}{10 * A} = \frac{343.35}{\pi * \frac{d^2}{4}} N$$

Shear force that each bolt needs to withstand:

$$\tau = \frac{T}{number\ of\ bolt * r * A} = \frac{16607.472}{10 * 0.4 * A} = \frac{4151.868}{\pi * \frac{d^2}{4}} N$$

By testing all bolt diameter and maximum normal and tensile strength from the table [8], M39 grade 10.9 bolt is chosen, both length is assumed to be 100mm

3 Results

3.1 Shaft arrangement

The upper guide bearing is positioned at the top of the shaft (0 meters). The rotor is placed 0.5 meters below the upper guide bearing, followed by the thrust bearing, which is positioned 0.5 meters below the rotor. The lower guide bearing is situated 0.5 meters below the thrust bearing and 1 meter above the turbine. This placement keeps the bearings away from water, which is crucial for components that require frequent lubrication. Additionally, maximizing the height of the water column enhances the efficiency of waterpower harnessing.

3.2 Component design result

Table 1: Shaft Design Result

Length	Diameter	Material	Safety factor	Net Weight
4.2m	119.8mm	AISI 316 Stainless steel	2	506.339kg

Table 2: Key - keyway Design Result

Length	Width	Depth
169mm	32mm	18.4mm

Table 3: Bolt Selection Result

Quantity	Length	Diameter	Model name
10	100mm	39mm	Grade 10.9

Table 4: Lower guide bearings selection result

Outer diameter	Bore diameter	Net weight	Dynamic load rating	Model Name

280mm	130mm	15.2kg	229kN	6326 Deep groove ball bearing

Table 5: Upper guide bearings selection result

Outer diameter	Bore diameter	Net weight	Dynamic load rating	Model name
200mm	130mm	2.31kg	83.2kN	16026 Deep groove ball bearing

Table 6: Thrust bearings selection result

Outer diameter	Bore diameter	Net weight	Dynamic load rating	Model name
225mm	130mm	8.79kg	865kN	29326 E Spherical roller thrust bearing with stamped steel cage

4. Discussion

4.1 Design implications:

The design ensures balanced load distribution, efficient torque transfer, and reliable operation over an 8-year period with a 98% reliability rate and a safety factor of 2.

4.2 Comparisons with standard:

Overall, the weight of the design is somewhat lower than the standard. By assuming the lowest weights for the turbine and rotor, this hydrogenerator is expected to be more cost-effective than the standard. However, due to the lack of precise measurements, the structural integrity of the design components may not match that of the standard.

4.3 Limitations:

This design assumes the best-case scenarios for weight, with the turbine and rotor weights taken as the lowest median values. Tolerances and clearances have not yet been specified, and all calculations are based on static loading conditions. However, the actual operating environment will involve dynamic conditions.

4.4 Recommendation:

While the design meets the specified requirements, further work with detailed simulations and material testing is recommended to check the assumptions and improve overall performance.

5. Conclusions

This project first calculates the shaft dimensions, then selects the appropriate bearing types. Next, the key and keyways are determined based on the shaft diameter, and finally, the bolts for the connections between the shaft and turbine are designed.

The project meets all requirements with the following specifications:

- Key width and length: 32mm and 169mm
- Bolt diameter and height: 39mm and 100mm
- Shaft diameter and length: 119.9mm and 4.2m

All bearings can withstand the applied load from the water and the weight of the generator. The bearings have an inner diameter of 130mm, which perfectly matches the shaft diameter. In summary, this project represents a significant step towards developing sustainable energy solutions using hydropower.

6. Appendix

6.1 Citation

[1] B. Bamason, "Guide to writing an engineering report," MIET2510 Mechanical Design. [Online]. Available: https://bamason2.github.io/miet2510-module/assessments/report_writing.html. [Accessed: Sep. 27, 2024].

[2] M. Akbar, "Material properties of AISI 316 stainless steel," ccitonlinewiki. [Online]. Available: https://air.eng.ui.ac.id/index.php?title=File:Material-Properties-of-AISI-316-Stainless-Steel-Abay.png. [Accessed: Sep. 27, 2024].

[3] Vedantu, "Water pressure formula - equation, calculation, examples and FAQs," [Online]. Available: https://www.vedantu.com/formula/water-pressure-formula. [Accessed: Sep. 27, 2024].

[4] B. Bamason, "MIET2510 Mechanical Design module," [Online]. Available: https://bamason2.github.io/miet2510-module/. [Accessed: Sep. 27, 2024].

[5] Desmos, "Desmos graphing calculator," [Online]. Available: https://www.desmos.com/calculator/ioc9fl3n2v?lang=vi. [Accessed: Sep. 27, 2024].

[6] SKF, "Deep groove ball bearing 6326," [Online]. Available: https://www.skf.com/us/products/rolling-bearings/ball-bearings/deep-groove-ball-bearings/productid-6326. [Accessed: Sep. 27, 2024].

[7] ISC Companies, "Keyway and key size dimensions," [Online]. Available: https://www.isccompanies.com/wp-content/uploads/2017/01/Keyway-and-Key-Size-Dimensions.pdf. [Accessed: Sep. 27, 2024].

[8] EurocodeApplied, "Bolt design properties," [Online]. Available: https://eurocodeapplied.com/design/en1993/bolt-design-properties. [Accessed: Sep. 27, 2024].

6.2 2D Drawing:

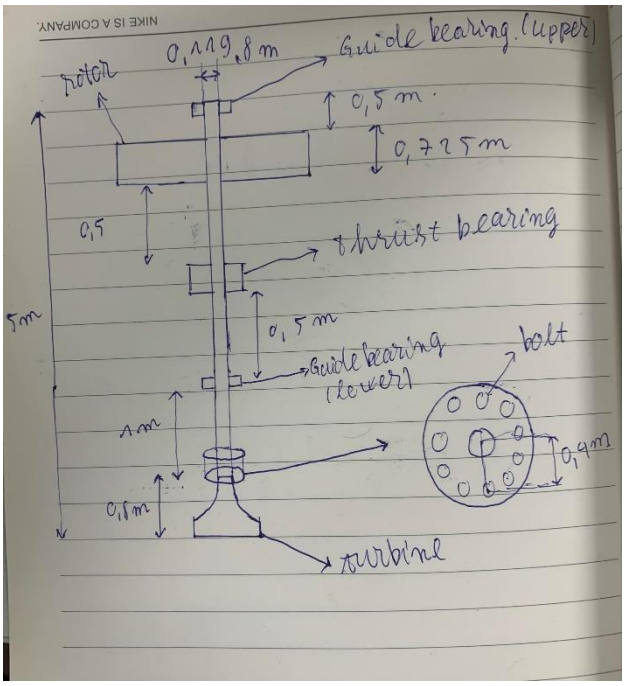


Figure 2: 2D drawing of final design for hydro generator